

## practice DQL statement

Write SQL statement for the following

1. To find all managers with salary >1500

```
mysql> select ename,mgr,sal from emp where sal>1500;
+-----+-----+-----+
| ename  | mgr  | sal      |
+-----+-----+-----+
| ALLEN  | 7698 | 1600.00  |
| JONES  | 7839 | 2975.00  |
| BLAKE  | 7839 | 2850.00  |
| jay_dbda | 7902 | 3000.00  |
| tanu_dbda | 7902 | 3000.00  |
| CLARK  | 7839 | 2450.00  |
| SCOTT  | 7566 | 3000.00  |
| KING   | NULL | 5000.00  |
| FORD   | 7566 | 3000.00  |
+-----+-----+-----+
9 rows in set (0.00 sec)
```

2. list all employees with sal >1200 and < 2000

```
mysql> select ename,sal from emp where sal>1200 and sal<2000;
+-----+-----+
| ename  | sal      |
+-----+-----+
| ALLEN  | 1600.00  |
| WARD   | 1250.00  |
| MARTIN | 1250.00  |
| TURNER | 1500.00  |
| MILLER | 1300.00  |
+-----+-----+
5 rows in set (0.00 sec)
```

2. list all employees with sal is 1600 or sal is 800 or sal is 1900

```
mysql> select ename,sal from emp where sal=1600 or sal=800 or sal=1900;
+-----+-----+
| ename  | sal      |
+-----+-----+
| SMITH  | 800.00   |
| ALLEN  | 1600.00  |
+-----+-----+
2 rows in set (0.00 sec)
```

4. list all employees with R at second last position in name

```
mysql> select ename from emp where ename like '%r_';
+-----+
| ename |
+-----+
| WARD  |
| CLARK |
| FORD  |
+-----+
3 rows in set (0.00 sec)

mysql> select ename from emp where ename regexp 'r.$';
+-----+
| ename |
+-----+
| WARD  |
| CLARK |
| FORD  |
+-----+
3 rows in set (0.00 sec)
```

5. List all employees with name starts with A and ends with N

```
mysql> select * from emp;
```

| EMPNO | ENAME     | JOB       | MGR  | HIREDATE   | SAL      | COMM    | DEPTNO |
|-------|-----------|-----------|------|------------|----------|---------|--------|
| 123   | Revati    | design    | NULL | 2020-03-14 | 55678.00 | 4567.00 | NULL   |
| 7369  | SMITH     | CLERK     | 7902 | 1980-12-17 | 800.00   | NULL    | 20     |
| 7499  | ALLEN     | SALESMAN  | 7698 | 1981-02-20 | 1600.00  | 300.00  | 30     |
| 7521  | WARD      | SALESMAN  | 7698 | 1981-02-22 | 1250.00  | 500.00  | 30     |
| 7566  | JONES     | MANAGER   | 7839 | 1981-04-02 | 2975.00  | NULL    | 20     |
| 7654  | MARTIN    | SALESMAN  | 7698 | 1981-09-28 | 1250.00  | 1400.00 | 30     |
| 7698  | BLAKE     | MANAGER   | 7839 | 1981-05-01 | 2850.00  | NULL    | 30     |
| 7780  | jay_dbda  | analyst   | 7902 | 2024-10-08 | 3000.00  | 300.00  | 20     |
| 7781  | tanu_dbda | analyst   | 7902 | 2024-10-08 | 3000.00  | 300.00  | 20     |
| 7782  | CLARK     | MANAGER   | 7839 | 1981-06-09 | 2450.00  | NULL    | 10     |
| 7788  | SCOTT     | ANALYST   | 7566 | 1982-12-09 | 3000.00  | NULL    | 20     |
| 7839  | KING      | PRESIDENT | NULL | 1981-11-17 | 5000.00  | NULL    | 10     |
| 7844  | TURNER    | SALESMAN  | 7698 | 1981-09-08 | 1500.00  | 0.00    | 30     |
| 7876  | ADAMS     | CLERK     | 7788 | 1983-01-12 | 1100.00  | NULL    | 20     |
| 7900  | JAMES     | CLERK     | 7698 | 1981-12-03 | 950.00   | NULL    | 30     |
| 7902  | FORD      | ANALYST   | 7566 | 1981-12-03 | 3000.00  | NULL    | 20     |
| 7934  | MILLER    | CLERK     | 7782 | 1982-01-23 | 1300.00  | NULL    | 10     |

```

17 rows in set (0.00 sec)

mysql> select ename from emp where ename like 'a%n';
+-----+
| ename |
+-----+
| ALLEN |
+-----+
1 row in set (0.00 sec)

mysql> select ename from emp where ename regexp '^a.*n$';
+-----+
| ename |
+-----+
| ALLEN |
+-----+
1 row in set (0.00 sec)

```

Q2. Solve following

1. list all employees with salary > 1250 and dept no=30

```
mysql> select sal,deptno,ename from emp where sal>1250 and deptno=30;
```

| sal     | deptno | ename  |
|---------|--------|--------|
| 1600.00 | 30     | ALLEN  |
| 2850.00 | 30     | BLAKE  |
| 1500.00 | 30     | TURNER |

```

3 rows in set (0.00 sec)

```

2. list all employees with salary  $\geq 1250$  and  $\leq 3000$

```
mysql> select sal,deptno,ename from emp where sal between 1250 and 3000;
```

| sal     | deptno | ename     |
|---------|--------|-----------|
| 1600.00 | 30     | ALLEN     |
| 1250.00 | 30     | WARD      |
| 2975.00 | 20     | JONES     |
| 1250.00 | 30     | MARTIN    |
| 2850.00 | 30     | BLAKE     |
| 3000.00 | 20     | jay_dbda  |
| 3000.00 | 20     | tanu_dbda |
| 2450.00 | 10     | CLARK     |
| 3000.00 | 20     | SCOTT     |
| 1500.00 | 30     | TURNER    |
| 3000.00 | 20     | FORD      |
| 1300.00 | 10     | MILLER    |

```
12 rows in set (0.00 sec)
```

3. list all employees with salary  $> 1250$  and  $< 3000$

```
mysql> select sal,deptno,ename from emp where sal>1250 and sal<3000;
```

| sal     | deptno | ename  |
|---------|--------|--------|
| 1600.00 | 30     | ALLEN  |
| 2975.00 | 20     | JONES  |
| 2850.00 | 30     | BLAKE  |
| 2450.00 | 10     | CLARK  |
| 1500.00 | 30     | TURNER |
| 1300.00 | 10     | MILLER |

```
6 rows in set (0.00 sec)
```

4. list all employees with salary either equal to 3000 or 1250 or 2500

```
mysql> select sal,deptno,ename from emp where sal=3000 or sal=1250 or sal=2500;
```

| sal     | deptno | ename     |
|---------|--------|-----------|
| 1250.00 | 30     | WARD      |
| 1250.00 | 30     | MARTIN    |
| 3000.00 | 20     | jay_dbda  |
| 3000.00 | 20     | tanu_dbda |
| 3000.00 | 20     | SCOTT     |
| 3000.00 | 20     | FORD      |

```
6 rows in set (0.00 sec)
```

5. list all employee with name=SMITH

```
mysql> select sal,deptno,ename from emp where ename='smith';
+-----+-----+-----+
| sal    | deptno | ename  |
+-----+-----+-----+
| 800.00 | 20     | SMITH  |
+-----+-----+-----+
1 row in set (0.00 sec)
```

6. list all employees with name starting with S

```
mysql> select sal,deptno,ename from emp where ename like 's%';
+-----+-----+-----+
| sal    | deptno | ename  |
+-----+-----+-----+
| 800.00 | 20     | SMITH  |
| 3000.00 | 20    | SCOTT  |
+-----+-----+-----+
2 rows in set (0.00 sec)
```

```
mysql> select sal,deptno,ename from emp where ename regexp '^s';
+-----+-----+-----+
| sal    | deptno | ename  |
+-----+-----+-----+
| 800.00 | 20     | SMITH  |
| 3000.00 | 20    | SCOTT  |
+-----+-----+-----+
2 rows in set (0.00 sec)
```

7. list all employees with name ending with S

```
mysql> select sal,deptno,ename from emp where ename like '%s';
+-----+-----+-----+
| sal    | deptno | ename  |
+-----+-----+-----+
| 2975.00 | 20     | JONES  |
| 1100.00 | 20     | ADAMS  |
| 950.00  | 30     | JAMES  |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> select sal,deptno,ename from emp where ename regexp 's$';
+-----+-----+-----+
| sal    | deptno | ename  |
+-----+-----+-----+
| 2975.00 | 20     | JONES  |
| 1100.00 | 20     | ADAMS  |
| 950.00  | 30     | JAMES  |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

8. list all employees with name contains l at 2nd position

```
mysql> select sal,deptno,ename from emp where ename regexp '^..l';
+-----+-----+-----+
| sal      | deptno | ename  |
+-----+-----+-----+
| 5000.00  | 10     | KING   |
| 1300.00  | 10     | MILLER |
+-----+-----+-----+
2 rows in set (0.00 sec)
```

9. list all employees with name starts with A ends with N and somewhere in between L is there

```
mysql> select sal,deptno,ename,job from emp where ename regexp '^a.*l.*n$';
+-----+-----+-----+-----+
| sal      | deptno | ename  | job      |
+-----+-----+-----+-----+
| 1600.00  | 30     | ALLEN  | SALESMAN |
+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

10. list all employees with name starts with A and B at 3 rd position and P at second last position

```
mysql> select sal,deptno,ename,job from emp where ename regexp '^t.n.*d.$';
+-----+-----+-----+-----+
| sal      | deptno | ename  | job      |
+-----+-----+-----+-----+
| 3000.00  | 20     | tanu_dbda | analyst |
+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> select sal,deptno,ename,job from emp where ename regexp '^a.b.*p.$';
Empty set (0.00 sec)
```

11. List all employees with name starts with either A or starts with S or starts with W

```
mysql> select sal,deptno,ename,job from emp where ename like 'a%' or ename like 's%' or ename like 'w%';
```

| sal     | deptno | ename | job      |
|---------|--------|-------|----------|
| 800.00  | 20     | SMITH | CLERK    |
| 1600.00 | 30     | ALLEN | SALESMAN |
| 1250.00 | 30     | WARD  | SALESMAN |
| 3000.00 | 20     | SCOTT | ANALYST  |
| 1100.00 | 20     | ADAMS | CLERK    |

```
5 rows in set (0.00 sec)
```

```
mysql> select sal,deptno,ename,job from emp where ename regexp '^[ASW]';
```

| sal     | deptno | ename | job      |
|---------|--------|-------|----------|
| 800.00  | 20     | SMITH | CLERK    |
| 1600.00 | 30     | ALLEN | SALESMAN |
| 1250.00 | 30     | WARD  | SALESMAN |
| 3000.00 | 20     | SCOTT | ANALYST  |
| 1100.00 | 20     | ADAMS | CLERK    |

```
5 rows in set (0.00 sec)
```

### practice Aggregate functions

12. find max sal and min sal for each job

```
mysql> select * from emp;
```

| EMPNO | ENAME     | JOB       | MGR  | HIREDATE   | SAL     | COMM    | DEPTNO |
|-------|-----------|-----------|------|------------|---------|---------|--------|
| 7369  | SMITH     | CLERK     | 7902 | 1980-12-17 | 800.00  | NULL    | 20     |
| 7499  | ALLEN     | SALESMAN  | 7698 | 1981-02-20 | 1600.00 | 300.00  | 30     |
| 7521  | WARD      | SALESMAN  | 7698 | 1981-02-22 | 1250.00 | 500.00  | 30     |
| 7566  | JONES     | MANAGER   | 7839 | 1981-04-02 | 2975.00 | NULL    | 20     |
| 7654  | MARTIN    | SALESMAN  | 7698 | 1981-09-28 | 1250.00 | 1400.00 | 30     |
| 7698  | BLAKE     | MANAGER   | 7839 | 1981-05-01 | 2850.00 | NULL    | 30     |
| 7780  | jay_dbda  | analyst   | 7902 | 2024-10-08 | 3000.00 | 300.00  | 20     |
| 7781  | tanu_dbda | analyst   | 7902 | 2024-10-08 | 3000.00 | 300.00  | 20     |
| 7782  | CLARK     | MANAGER   | 7839 | 1981-06-09 | 2450.00 | NULL    | 10     |
| 7788  | SCOTT     | ANALYST   | 7566 | 1982-12-09 | 3000.00 | NULL    | 20     |
| 7839  | KING      | PRESIDENT | NULL | 1981-11-17 | 5000.00 | NULL    | 10     |
| 7844  | TURNER    | SALESMAN  | 7698 | 1981-09-08 | 1500.00 | 0.00    | 30     |
| 7876  | ADAMS     | CLERK     | 7788 | 1983-01-12 | 1100.00 | NULL    | 20     |
| 7900  | JAMES     | CLERK     | 7698 | 1981-12-03 | 950.00  | NULL    | 30     |
| 7902  | FORD      | ANALYST   | 7566 | 1981-12-03 | 3000.00 | NULL    | 20     |
| 7934  | MILLER    | CLERK     | 7782 | 1982-01-23 | 1300.00 | NULL    | 10     |

```
16 rows in set (0.00 sec)
```

```
mysql> select max(sal),min(sal) from emp;
```

| max(sal) | min(sal) |
|----------|----------|
| 5000.00  | 800.00   |

```
1 row in set (0.00 sec)
```

13. find how many employees have not received commission

```
mysql> select ename,comm,sal from emp where comm is null or comm=0;
+-----+-----+-----+
| ename  | comm  | sal    |
+-----+-----+-----+
| SMITH  | NULL  | 800.00 |
| JONES  | NULL  | 2975.00|
| BLAKE  | NULL  | 2850.00|
| CLARK  | NULL  | 2450.00|
| SCOTT  | NULL  | 3000.00|
| KING   | NULL  | 5000.00|
| TURNER | 0.00  | 1500.00|
| ADAMS  | NULL  | 1100.00|
| JAMES  | NULL  | 950.00 |
| FORD   | NULL  | 3000.00|
| MILLER | NULL  | 1300.00|
+-----+-----+-----+
11 rows in set (0.00 sec)
```

14. find sum of sal of all employees working in dept no 10

```
mysql> select sum(sal) 'Sum_Sal' from emp where deptno=10;
+-----+
| Sum_Sal |
+-----+
| 8750.00 |
+-----+
1 row in set (0.00 sec)
```

15. find maximum salary,average sal for each job in every department

```
mysql> select sum(sal) 'Sum_Sal',avg(sal) 'Avg_Sal',max(sal) 'Max_Sal' from emp;
+-----+-----+-----+
| Sum_Sal | Avg_Sal      | Max_Sal |
+-----+-----+-----+
| 35025.00 | 2189.062500 | 5000.00 |
+-----+-----+-----+
1 row in set (0.00 sec)
```

16. find max salary for every department if deptno is > 15 and arrange data in deptno order.

```
mysql> select deptno,max(sal) 'Sal' from emp where deptno>15 group by deptno order by deptno;
+-----+-----+
| deptno | Sal      |
+-----+-----+
|      20 | 3000.00 |
|      30 | 2850.00 |
+-----+-----+
2 rows in set (0.00 sec)
```

17. find sum salary for every department if sum is > 3000

```
mysql> select deptno,sum(sal) from emp group by deptno having sum(sal)>3000;
+-----+-----+
| deptno | sum(sal) |
+-----+-----+
|      44 | 12345.00 |
|      20 | 80093.75 |
|      30 | 10200.00 |
|      10 | 14312.50 |
+-----+-----+
4 rows in set (0.00 sec)
```

18. list all department which has minimum 5 employees

```
mysql> select count(*),deptno from emp group by deptno having count(*)=5;
+-----+-----+
| count(*) | deptno |
+-----+-----+
|         5 |      20 |
+-----+-----+
1 row in set (0.00 sec)
```

19. count how many employees earn salary more than 2000 in each job

```
mysql> select count(sal) from emp where sal>2000;
+-----+
| count(sal) |
+-----+
|           6 |
+-----+
1 row in set (0.10 sec)
```

20. list all enames and jobs in small case letter

```
mysql> select lower(ename),lower(job) from emp;
+-----+-----+
| lower(ename) | lower(job) |
+-----+-----+
| smith       | clerk      |
| allen       | salesman   |
| ward        | salesman   |
| jones       | manager    |
| martin      | salesman   |
| blake       | manager    |
| clark       | manager    |
| scott       | analyst    |
| king        | president  |
| turner      | salesman   |
| adams       | clerk      |
| james       | clerk      |
| ford        | analyst    |
| miller      | clerk      |
+-----+-----+
14 rows in set (0.01 sec)
```

21. list all names and jobs so that the length of name should be 15 if it is smaller then add spaces to left

```
mysql> SELECT LPAD(ename,15,'**') from emp;
```

| LPAD(ename,15,'**') |
|---------------------|
| *****SMITH          |
| *****ALLEN          |
| *****WARD           |
| *****JONES          |
| *****MARTIN         |
| *****BLAKE          |
| *****CLARK          |
| *****SCOTT          |
| *****KING           |
| *****TURNER         |
| *****ADAMS          |
| *****JAMES          |
| *****FORD           |
| *****MILLER         |

```
14 rows in set (0.00 sec)
```

22. display min sal,max sal, average sal for all employees working under same manager

```
mysql> select min(sal),max(sal),avg(sal) from emp group by mgr;
```

| min(sal) | max(sal) | avg(sal)    |
|----------|----------|-------------|
| 800.00   | 3000.00  | 2266.666667 |
| 950.00   | 1600.00  | 1310.000000 |
| 2450.00  | 2975.00  | 2758.333333 |
| 3000.00  | 3000.00  | 3000.000000 |
| 5000.00  | 5000.00  | 5000.000000 |
| 1100.00  | 1100.00  | 1100.000000 |
| 1300.00  | 1300.00  | 1300.000000 |

```
7 rows in set (0.00 sec)
```

23. find sum of total earnings(sal+comm), average of sal+comm,  
for all employees who earn sal > 2000 and work in either dept no 10 or 20

```

mysql> select sum(sal+ifnull(comm,0)),avg(sal+ifnull(comm,0)) from emp where sal>2000 and deptno
in(10,20);
+-----+-----+
| sum(sal+ifnull(comm,0)) | avg(sal+ifnull(comm,0)) |
+-----+-----+
|                23025.00 |          3289.285714 |
+-----+-----+
1 row in set (0.00 sec)

```

24. list all employees who joined in Aug 1980 and salary is >1500 and < 2500

```

mysql> select ename,hiredate from emp where hiredate regexp '1980-08' and sal>1500 and sal<2500;
Empty set (0.00 sec)

```

25. list all employees joined in either aug or may or dec

```

mysql> select ename,hiredate from emp where month(hiredate)=08 or month(hi
redate)=05 or month(hiredate)=12;
+-----+-----+
| ename | hiredate |
+-----+-----+
| SMITH | 1980-12-17 |
| BLAKE | 1981-05-01 |
| SCOTT | 1982-12-09 |
| JAMES | 1981-12-03 |
| FORD  | 1981-12-03 |
+-----+-----+
5 rows in set (0.01 sec)

```

26. display name and hiredate in dd/mm/yy format for all employees whose job is clerk and they earn some commission

```

mysql> select empno,ename,hiredate,date_format(hiredate,'%d/%m/%Y') from emp where job='clerk' and
comm is not null;
Empty set (0.00 sec)

mysql> select empno,ename,hiredate,date_format(hiredate,'%d/%m/%Y') from emp;
+-----+-----+-----+-----+
| empno | ename   | hiredate | date_format(hiredate,'%d/%m/%Y') |
+-----+-----+-----+-----+
| 7369 | SMITH   | 1980-12-17 | 17/12/1980 |
| 7499 | ALLEN   | 1981-02-20 | 20/02/1981 |
| 7521 | WARD    | 1981-02-22 | 22/02/1981 |
| 7566 | JONES   | 1981-04-02 | 02/04/1981 |
| 7654 | MARTIN  | 1981-09-28 | 28/09/1981 |
| 7698 | BLAKE   | 1981-05-01 | 01/05/1981 |
| 7780 | jay_dbda | 2024-10-08 | 08/10/2024 |
| 7781 | tanu_dbda | 2024-10-08 | 08/10/2024 |
| 7782 | CLARK   | 1981-06-09 | 09/06/1981 |
| 7788 | SCOTT   | 1982-12-09 | 09/12/1982 |
| 7839 | KING    | 1981-11-17 | 17/11/1981 |
| 7844 | TURNER  | 1981-09-08 | 08/09/1981 |
| 7876 | ADAMS   | 1983-01-12 | 12/01/1983 |
| 7900 | JAMES   | 1981-12-03 | 03/12/1981 |
| 7902 | FORD    | 1981-12-03 | 03/12/1981 |
| 7934 | MILLER  | 1982-01-23 | 23/01/1982 |
+-----+-----+-----+-----+
16 rows in set (0.00 sec)

```

27. list empcode,empno,name and job for each employee.

(note :empcode is 3 to 5 characters from name and last 2 characters of job)

```
mysql> select concat(substr(ename,3,5),substr(job,1,2))empcode,empno,ename,job from emp;
```

| empcode | empno | ename     | job       |
|---------|-------|-----------|-----------|
| ITHCL   | 7369  | SMITH     | CLERK     |
| LENSA   | 7499  | ALLEN     | SALESMAN  |
| RDSA    | 7521  | WARD      | SALESMAN  |
| NESMA   | 7566  | JONES     | MANAGER   |
| RTINSA  | 7654  | MARTIN    | SALESMAN  |
| AKEMA   | 7698  | BLAKE     | MANAGER   |
| y_dbdan | 7780  | jay_dbda  | analyst   |
| nu_dban | 7781  | tanu_dbda | analyst   |
| ARKMA   | 7782  | CLARK     | MANAGER   |
| OTTAN   | 7788  | SCOTT     | ANALYST   |
| NGPR    | 7839  | KING      | PRESIDENT |
| RNERSA  | 7844  | TURNER    | SALESMAN  |
| AMSCL   | 7876  | ADAMS     | CLERK     |
| MESCL   | 7900  | JAMES     | CLERK     |
| RDAN    | 7902  | FORD      | ANALYST   |
| LLERCL  | 7934  | MILLER    | CLERK     |

```
16 rows in set (0.00 sec)
```

28. display thousand separator and \$ symbol for commission if it is null then display it as 0 for all employees whose name starts with A and ends with N

29. Display empid,name,sal,comm,remark Remark should base on following conditions

comm >= 600 "excellent Keep it up"

if it < 600 or not null "good"

otherwise "Need improvement"

```
mysql> SELECT EMPNO,ENAME,SAL,COMM,
-> CASE
-> WHEN COMM>600 THEN 'EXCELLENT KEEP IT UP'
-> WHEN COMM<600 THEN 'GOOD'
-> ELSE 'NEED IMPROVEMENT'
-> END AS COMM
-> FROM EMP;
```

| EMPNO | ENAME     | SAL     | COMM    | COMM                 |
|-------|-----------|---------|---------|----------------------|
| 7369  | SMITH     | 800.00  | NULL    | NEED IMPROVEMENT     |
| 7499  | ALLEN     | 1600.00 | 300.00  | GOOD                 |
| 7521  | WARD      | 1250.00 | 500.00  | GOOD                 |
| 7566  | JONES     | 2975.00 | NULL    | NEED IMPROVEMENT     |
| 7654  | MARTIN    | 1250.00 | 1400.00 | EXCELLENT KEEP IT UP |
| 7698  | BLAKE     | 2850.00 | NULL    | NEED IMPROVEMENT     |
| 7780  | jay_dbda  | 3000.00 | 300.00  | GOOD                 |
| 7781  | tanu_dbda | 3000.00 | 300.00  | GOOD                 |
| 7782  | CLARK     | 2450.00 | NULL    | NEED IMPROVEMENT     |
| 7788  | SCOTT     | 3000.00 | NULL    | NEED IMPROVEMENT     |
| 7839  | KING      | 5000.00 | NULL    | NEED IMPROVEMENT     |
| 7844  | TURNER    | 1500.00 | 0.00    | GOOD                 |
| 7876  | ADAMS     | 1100.00 | NULL    | NEED IMPROVEMENT     |
| 7900  | JAMES     | 950.00  | NULL    | NEED IMPROVEMENT     |
| 7902  | FORD      | 3000.00 | NULL    | NEED IMPROVEMENT     |
| 7934  | MILLER    | 1300.00 | NULL    | NEED IMPROVEMENT     |

16 rows in set (0.00 sec)

30. Display empid, name, deptno and department name by using  
following conditions.

dept 10 then "Hr"

if 20 then "Admin"

if 30 then "accounts"

otherwise purchase

```
mysql>
mysql> SELECT EMPNO,ENAME,DEPTNO,
-> CASE
-> WHEN DEPTNO=10 THEN 'HR'
-> WHEN DEPTNO=20 THEN 'ADMIN'
-> WHEN DEPTNO=30 THEN 'ACCONTS'
-> ELSE 'PURCHASE'
-> END AS DEPTNO
-> FROM EMP;
```

| EMPNO | ENAME     | DEPTNO | DEPTNO  |
|-------|-----------|--------|---------|
| 7369  | SMITH     | 20     | ADMIN   |
| 7499  | ALLEN     | 30     | ACCONTS |
| 7521  | WARD      | 30     | ACCONTS |
| 7566  | JONES     | 20     | ADMIN   |
| 7654  | MARTIN    | 30     | ACCONTS |
| 7698  | BLAKE     | 30     | ACCONTS |
| 7780  | jay_dbda  | 20     | ADMIN   |
| 7781  | tanu_dbda | 20     | ADMIN   |
| 7782  | CLARK     | 10     | HR      |
| 7788  | SCOTT     | 20     | ADMIN   |
| 7839  | KING      | 10     | HR      |
| 7844  | TURNER    | 30     | ACCONTS |
| 7876  | ADAMS     | 20     | ADMIN   |
| 7900  | JAMES     | 30     | ACCONTS |
| 7902  | FORD      | 20     | ADMIN   |
| 7934  | MILLER    | 10     | HR      |

```
16 rows in set (0.00 sec)
```

Topic ----- create Table, DML , subquery and joins

31. Practice creating following tables

```
create table mydept_DBDA
(
deptid int primary key,
dname varchar(20) not null unique,
dloc varchar(20)
)
```

```
mysql>
mysql> CREATE TABLE MYDEPT_DBDA
-> (
-> DEPTID INT PRIMARY KEY,
-> DNAME VARCHAR(20) NOT NULL UNIQUE,
-> DLOC VARCHAR(20)
-> )
-> ;
Query OK, 0 rows affected (0.84 sec)

mysql> SELECT * FROM MYDEPT_DBDA;
Empty set (0.00 sec)

mysql> SHOW TABLES;
+-----+
| Tables_in_iacsd0324 |
+-----+
| bonus                |
| customers            |
| dept                 |
| emp                  |
| medicine             |
| mydept_dbda         |
| orders               |
| player              |
| salgrade             |
| suppliers            |
| team                 |
+-----+
11 rows in set (0.00 sec)
```

Oracle syntax:

```
create table mydept_DBDA
```

```
(  
deptid number primary key,  
dname varchar2(20) not null unique,  
dloc varchar2(20)  
)  
insert into mydept_DBDA values(30,'Purchase','Mumbai');
```

MySql syntax:

```
create table myemployee1  
(  
empno int primary key,  
fname varchar(15) not null,  
mname varchar(15),  
lname varchar(15) not null,  
sal float(9,2) check(sal >=1000),  
doj date,  
passportnum varchar(15) unique,  
deptno int,  
constraint fk_deptno foreign key(deptno) references mydept_DBDA(deptid) on  
delete set null  
on update cascade  
)
```

```

mysql> CREATE TABLE MYEMPLOYEE(
  -> EMPNO INT PRIMARY KEY,
  -> FNAME VARCHAR(15) NOT NULL,
  -> MNAME VARCHAR(15),
  -> LNAME VARCHAR(15) NOT NULL ,
  -> SAL FLOAT(9,2) CHECK (SAL>=1000),
  -> DOJ DATE,
  -> PASSPORTNUM VARCHAR(15) UNIQUE,
  -> DEPTNO INT,
  -> CONSTRAINT FK_DEPTNO FOREIGN KEY(DEPTNO) REFERENCES MYDEPT_DBDA(DEPTID) ON DELETE SET NULL
  -> ON UPDATE CASCADE);
Query OK, 0 rows affected, 1 warning (1.00 sec)

mysql> SHOW TABLES;
+-----+
| Tables_in_iacsd0324 |
+-----+
| bonus                |
| customers            |
| dept                 |
| emp                  |
| medicine             |
| mydept_dbda          |
| myemployee           |
| orders               |
| player               |
| salgrade             |
| suppliers            |
| team                 |
+-----+
12 rows in set (0.00 sec)

```

Oracle syntax:

create table myemployee

(

empno number(5) primary key,

fname varchar2(15) not null,

mname varchar2(15),

lname varchar2(15) not null,

sal number(9,2) check(sal >=1000),

doj date default sysdate,

passportnum varchar2(15) unique,

deptno number constraint fk\_deptno references mydept\_DBDA(deptid) on delete cascade

)

32. Create following tables Student, Course

Student (sid,sname) ----- sid ---primary key

```
mysql> CREATE TABLE STUDENT(  
-> SID INT PRIMARY KEY,  
-> SNAME VARCHAR(20));  
Query OK, 0 rows affected (0.85 sec)  
  
mysql> SHOW TABLES;  
+-----+  
| Tables_in_iacsd0324 |  
+-----+  
| bonus  
| customers  
| dept  
| emp  
| medicine  
| mydept_dbda  
| myemployee  
| orders  
| player  
| salgrade  
| student  
| suppliers  
| team  
+-----+  
13 rows in set (0.00 sec)
```

Course(cid,cname)----- cid ---primary key

```
mysql> CREATE TABLE COURSE(  
-> CID INT PRIMARY KEY,  
-> CNAME VARCHAR(20));  
Query OK, 0 rows affected (0.73 sec)
```

```
mysql> SHOW TABLES;
```

```
+-----+  
| Tables_in_iacsd0324 |  
+-----+  
| bonus                |  
| course               |  
| customers            |  
| dept                 |  
| emp                  |  
| medicine             |  
| mydept_dbda          |  
| myemployee           |  
| orders               |  
| player               |  
| salgrade             |  
| student              |  
| suppliers            |  
| team                 |  
+-----+
```

```
14 rows in set (0.00 sec)
```

Marks(studid,courseid,marks)

```
mysql> CREATE TABLE MARKS(
-> STUDID INT PRIMARY KEY,
-> COURSEID INT,
-> MARKS INT );
Query OK, 0 rows affected (0.32 sec)

mysql> SHOW TABLES;
+-----+
| Tables_in_iacsd0324 |
+-----+
| bonus                |
| course               |
| customers            |
| dept                |
| emp                 |
| marks               |
| medicine             |
| mydept_dbda         |
| myemployee          |
| orders              |
| player              |
| salgrade            |
| student             |
| suppliers            |
| team                |
+-----+
15 rows in set (0.00 sec)
```

Sample data for marks table

studid,courseid,maks

1 1 99

1 3 98

2 1 95

2 2 97

create table marks(

studid number,

courseid number,

marks number,

constraint pk primary key(studid,courseid),

constraint fk\_sid foreign key (studid) references student(sid) on delete cascade,

constraint fk\_cid foreign key (courseid) references course(cid)

)

```
constraint pk primary key (studid,courseid) on delete cascade,
mysql> create table marks(
  -> studid int,
  -> courseid int,
  -> marks int,
  -> constraint pk primary key (studid,courseid),
  -> constraint fk_sid foreign key (studid) references student(sid) on delete cascade, constraint fk_cid
foreign key (courseid) references course(cid)
  -> );
```

Query OK, 0 rows affected (0.39 sec)

mysql> show tables

-> ;

```
+-----+
| Tables_in_iacsd0324 |
+-----+
```

```
| bonus
| course
| customers
| dept
| emp
| marks
| medicine
| mydept_dbda
| myemployee
| orders
| player
| salgrade
| student
| suppliers
| team
+-----+
```

15 rows in set (0.00 sec)

33. Create empty table emp10 with table structure same as emp table.

create table tanu as

(

select \*

from emp

where 1=2;

)

```

mysql> create table emp10 (
-> select * from emp
-> where 1=2)
-> ;
Query OK, 0 rows affected (0.29 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> select * from emp10
-> ;
Empty set (0.00 sec)

mysql> show tables;
+-----+
| Tables_in_iacsd0324 |
+-----+
| bonus                |
| course               |
| customers             |
| dept                  |
| emp                   |
| emp10                 |
| marks                 |
| medicine              |
| mydept_dbda           |
| myemployee            |
| orders                |
| player                |
| prac                  |
| salgrade              |
| student               |
| suppliers              |
| team                  |
+-----+
17 rows in set (0.00 sec)

```

34. Solve following using alter table

add primary key constraint on emp,dept,salgrade

emp ----> empno

dept---> deptno

salgrade---> grade

add foreign key constarint in emp

deptno --->> dept(deptno)

add new column in emp table netsal with constraint default 1000

35. Update employee sal ---- increase sal of each employee by 15 % sal +comm, change the job to manager and mgr to 7777 for all employees in deptno 10.

36. change job of smith to senior clerk

37. increase salary of all employees by 15% if they are earning some commission

38. list all employees with sal>smith's sal

```
mysql> select * from emp where sal>(select sal from emp where ename='smith');
+-----+-----+-----+-----+-----+-----+-----+-----+
| EMPNO | ENAME   | JOB       | MGR  | HIREDATE | SAL      | COMM    | DEPTNO |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 7499 | ALLEN   | SALESMAN  | 7698 | 1981-02-20 | 1600.00 | 300.00 | 30      |
| 7521 | WARD    | SALESMAN  | 7698 | 1981-02-22 | 1250.00 | 500.00 | 30      |
| 7566 | JONES   | MANAGER   | 7839 | 1981-04-02 | 2975.00 | NULL    | 20      |
| 7654 | MARTIN  | SALESMAN  | 7698 | 1981-09-28 | 1250.00 | 1400.00 | 30      |
| 7698 | BLAKE   | MANAGER   | 7839 | 1981-05-01 | 2850.00 | NULL    | 30      |
| 7780 | jay_dbda | analyst   | 7902 | 2024-10-08 | 3000.00 | 300.00 | 20      |
| 7781 | tanu_dbda | analyst   | 7902 | 2024-10-08 | 3000.00 | 300.00 | 20      |
| 7782 | CLARK   | MANAGER   | 7839 | 1981-06-09 | 2450.00 | NULL    | 10      |
| 7788 | SCOTT   | ANALYST   | 7566 | 1982-12-09 | 3000.00 | NULL    | 20      |
| 7839 | KING    | PRESIDENT | NULL | 1981-11-17 | 5000.00 | NULL    | 10      |
| 7844 | TURNER  | SALESMAN  | 7698 | 1981-09-08 | 1500.00 | 0.00    | 30      |
| 7876 | ADAMS   | CLERK     | 7788 | 1983-01-12 | 1100.00 | NULL    | 20      |
| 7900 | JAMES   | CLERK     | 7698 | 1981-12-03 | 950.00  | NULL    | 30      |
| 7902 | FORD    | ANALYST   | 7566 | 1981-12-03 | 3000.00 | NULL    | 20      |
| 7934 | MILLER  | CLERK     | 7782 | 1982-01-23 | 1300.00 | NULL    | 10      |
+-----+-----+-----+-----+-----+-----+-----+-----+
15 rows in set (0.00 sec)
```

39. list all employees who are working in smith's department

```
mysql> select * from emp
-> where deptno>(select deptno from emp where ename='smith');
+-----+-----+-----+-----+-----+-----+-----+-----+
| EMPNO | ENAME   | JOB       | MGR  | HIREDATE | SAL      | COMM    | DEPTNO |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 7499 | ALLEN   | SALESMAN  | 7698 | 1981-02-20 | 2433.40 | 300.00 | 30      |
| 7521 | WARD    | SALESMAN  | 7698 | 1981-02-22 | 1901.10 | 500.00 | 30      |
| 7654 | MARTIN  | SALESMAN  | 7698 | 1981-09-28 | 1901.10 | 1400.00 | 30      |
| 7698 | BLAKE   | MANAGER   | 7839 | 1981-05-01 | 2850.00 | NULL    | 30      |
| 7844 | TURNER  | SALESMAN  | 7698 | 1981-09-08 | 1500.00 | 0.00    | 30      |
| 7900 | JAMES   | CLERK     | 7698 | 1981-12-03 | 950.00  | NULL    | 30      |
+-----+-----+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)
```

```
mysql> select * from emp;
```

| EMPNO | ENAME     | JOB       | MGR  | HIREDATE   | SAL     | COMM    | DEPTNO |
|-------|-----------|-----------|------|------------|---------|---------|--------|
| 7369  | SMITH     | CLERK     | 7902 | 1980-12-17 | 800.00  | NULL    | 20     |
| 7499  | ALLEN     | SALESMAN  | 7698 | 1981-02-20 | 1600.00 | 300.00  | 30     |
| 7521  | WARD      | SALESMAN  | 7698 | 1981-02-22 | 1250.00 | 500.00  | 30     |
| 7566  | JONES     | MANAGER   | 7839 | 1981-04-02 | 2975.00 | NULL    | 20     |
| 7654  | MARTIN    | SALESMAN  | 7698 | 1981-09-28 | 1250.00 | 1400.00 | 30     |
| 7698  | BLAKE     | MANAGER   | 7839 | 1981-05-01 | 2850.00 | NULL    | 30     |
| 7780  | jay_dbda  | analyst   | 7902 | 2024-10-08 | 3000.00 | 300.00  | 20     |
| 7781  | tanu_dbda | analyst   | 7902 | 2024-10-08 | 3000.00 | 300.00  | 20     |
| 7782  | CLARK     | MANAGER   | 7839 | 1981-06-09 | 2450.00 | NULL    | 10     |
| 7788  | SCOTT     | ANALYST   | 7566 | 1982-12-09 | 3000.00 | NULL    | 20     |
| 7839  | KING      | PRESIDENT | NULL | 1981-11-17 | 5000.00 | NULL    | 10     |
| 7844  | TURNER    | SALESMAN  | 7698 | 1981-09-08 | 1500.00 | 0.00    | 30     |
| 7876  | ADAMS     | CLERK     | 7788 | 1983-01-12 | 1100.00 | NULL    | 20     |
| 7900  | JAMES     | CLERK     | 7698 | 1981-12-03 | 950.00  | NULL    | 30     |
| 7902  | FORD      | ANALYST   | 7566 | 1981-12-03 | 3000.00 | NULL    | 20     |
| 7934  | MILLER    | CLERK     | 7782 | 1982-01-23 | 1300.00 | NULL    | 10     |

```
16 rows in set (0.00 sec)
```

```
mysql> select ename from emp where deptno=(select deptno from emp where ename='smith');
```

| ename     |
|-----------|
| SMITH     |
| JONES     |
| jay_dbda  |
| tanu_dbda |
| SCOTT     |
| ADAMS     |
| FORD      |

```
7 rows in set (0.00 sec)
```

40. list all employees with sal < rajan's sal and salary > revati's sal

```
mysql> select * from emp
-> where sal>(select sal from emp where ename='allen') and sal<(select sal from emp where ename='scott');
```

| EMPNO | ENAME | JOB     | MGR  | HIREDATE   | SAL     | COMM | DEPTNO |
|-------|-------|---------|------|------------|---------|------|--------|
| 7566  | JONES | MANAGER | 7839 | 1981-04-02 | 2975.00 | NULL | 20     |
| 7698  | BLAKE | MANAGER | 7839 | 1981-05-01 | 2850.00 | NULL | 30     |
| 7782  | CLARK | MANAGER | 7839 | 1981-06-09 | 2450.00 | NULL | 10     |

```
3 rows in set (0.00 sec)
```

41. delete all employees working in alan's department

```
mysql> delete from emp
-> where deptno=(select deptno from (select* from emp) e where e.ename='allen');
Query OK, 6 rows affected (0.09 sec)
```

```
mysql> select * from emp;
```

| EMPNO | ENAME     | JOB       | MGR  | HIREDATE   | SAL     | COMM   | DEPTNO |
|-------|-----------|-----------|------|------------|---------|--------|--------|
| 7369  | SMITH     | CLERK     | 7902 | 1980-12-17 | 800.00  | NULL   | 20     |
| 7566  | JONES     | MANAGER   | 7839 | 1981-04-02 | 2975.00 | NULL   | 20     |
| 7780  | jay_dbda  | analyst   | 7902 | 2024-10-08 | 3000.00 | 300.00 | 20     |
| 7781  | tanu_dbda | analyst   | 7902 | 2024-10-08 | 3000.00 | 300.00 | 20     |
| 7782  | CLARK     | MANAGER   | 7839 | 1981-06-09 | 2450.00 | NULL   | 10     |
| 7788  | SCOTT     | ANALYST   | 7566 | 1982-12-09 | 3000.00 | NULL   | 20     |
| 7839  | KING      | PRESIDENT | NULL | 1981-11-17 | 5000.00 | NULL   | 10     |
| 7876  | ADAMS     | CLERK     | 7788 | 1983-01-12 | 1100.00 | NULL   | 20     |
| 7902  | FORD      | ANALYST   | 7566 | 1981-12-03 | 3000.00 | NULL   | 20     |
| 7934  | MILLER    | CLERK     | 7782 | 1982-01-23 | 1300.00 | NULL   | 10     |

```
10 rows in set (0.00 sec)
```

42. change salary of Alan to the salary of Miller.

```
mysql> update emp
-> set sal=(select sal from(select * from emp)e where ename='miller')
-> where ename='allen';
```

```
Query OK, 1 row affected (0.05 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql> select * from emp;
```

| EMPNO | ENAME     | JOB       | MGR  | HIREDATE   | SAL     | COMM    | DEPTNO |
|-------|-----------|-----------|------|------------|---------|---------|--------|
| 7369  | SMITH     | CLERK     | 7902 | 1980-12-17 | 800.00  | NULL    | 20     |
| 7499  | ALLEN     | SALESMAN  | 7698 | 1981-02-20 | 1300.00 | 300.00  | 30     |
| 7521  | WARD      | SALESMAN  | 7698 | 1981-02-22 | 1250.00 | 500.00  | 30     |
| 7566  | JONES     | MANAGER   | 7839 | 1981-04-02 | 2975.00 | NULL    | 20     |
| 7654  | MARTIN    | SALESMAN  | 7698 | 1981-09-28 | 1250.00 | 1400.00 | 30     |
| 7698  | BLAKE     | MANAGER   | 7839 | 1981-05-01 | 2850.00 | NULL    | 30     |
| 7780  | jay_dbda  | analyst   | 7902 | 2024-10-08 | 3000.00 | 300.00  | 20     |
| 7781  | tanu_dbda | analyst   | 7902 | 2024-10-08 | 3000.00 | 300.00  | 20     |
| 7782  | CLARK     | MANAGER   | 7839 | 1981-06-09 | 2450.00 | NULL    | 10     |
| 7788  | SCOTT     | ANALYST   | 7566 | 1982-12-09 | 3000.00 | NULL    | 20     |
| 7839  | KING      | PRESIDENT | NULL | 1981-11-17 | 5000.00 | NULL    | 10     |
| 7844  | TURNER    | SALESMAN  | 7698 | 1981-09-08 | 1500.00 | 0.00    | 30     |
| 7876  | ADAMS     | CLERK     | 7788 | 1983-01-12 | 1100.00 | NULL    | 20     |
| 7900  | JAMES     | CLERK     | 7698 | 1981-12-03 | 950.00  | NULL    | 30     |
| 7902  | FORD      | ANALYST   | 7566 | 1981-12-03 | 3000.00 | NULL    | 20     |
| 7934  | MILLER    | CLERK     | 7782 | 1982-01-23 | 1300.00 | NULL    | 10     |

```
16 rows in set (0.00 sec)
```

43. change salary of all employees who working in Wall's department to the salary of Miller.

44. list all employees with salary > either Smith's salary or alan's sal

45. list all employees who earn more than average sal of dept 10

```
mysql> select * from emp where sal > (select avg(sal) from emp where deptno=10);
```

| EMPNO | ENAME     | JOB       | MGR  | HIREDATE   | SAL     | COMM   | DEPTNO |
|-------|-----------|-----------|------|------------|---------|--------|--------|
| 7566  | JONES     | MANAGER   | 7839 | 1981-04-02 | 2975.00 | NULL   | 20     |
| 7780  | jay_dbda  | analyst   | 7902 | 2024-10-08 | 3000.00 | 300.00 | 20     |
| 7781  | tanu_dbda | analyst   | 7902 | 2024-10-08 | 3000.00 | 300.00 | 20     |
| 7788  | SCOTT     | ANALYST   | 7566 | 1982-12-09 | 3000.00 | NULL   | 20     |
| 7902  | FORD      | ANALYST   | 7566 | 1981-12-03 | 3000.00 | NULL   | 20     |
| 7839  | KING      | PRESIDENT | NULL | 1981-11-17 | 5000.00 | NULL   | 10     |

```
6 rows in set (0.00 sec)
```

46. list all employees who earn more than average sal of Alan's department

```
mysql> select * from emp;
```

| EMPNO | ENAME     | JOB       | MGR  | HIREDATE   | SAL     | COMM    | DEPTNO |
|-------|-----------|-----------|------|------------|---------|---------|--------|
| 7369  | SMITH     | CLERK     | 7902 | 1980-12-17 | 800.00  | NULL    | 20     |
| 7499  | ALLEN     | SALESMAN  | 7698 | 1981-02-20 | 1600.00 | 300.00  | 30     |
| 7521  | WARD      | SALESMAN  | 7698 | 1981-02-22 | 1250.00 | 500.00  | 30     |
| 7566  | JONES     | MANAGER   | 7839 | 1981-04-02 | 2975.00 | NULL    | 20     |
| 7654  | MARTIN    | SALESMAN  | 7698 | 1981-09-28 | 1250.00 | 1400.00 | 30     |
| 7698  | BLAKE     | MANAGER   | 7839 | 1981-05-01 | 2850.00 | NULL    | 30     |
| 7780  | jay_dbda  | analyst   | 7902 | 2024-10-08 | 3000.00 | 300.00  | 20     |
| 7781  | tanu_dbda | analyst   | 7902 | 2024-10-08 | 3000.00 | 300.00  | 20     |
| 7782  | CLARK     | MANAGER   | 7839 | 1981-06-09 | 2450.00 | NULL    | 10     |
| 7788  | SCOTT     | ANALYST   | 7566 | 1982-12-09 | 3000.00 | NULL    | 20     |
| 7839  | KING      | PRESIDENT | NULL | 1981-11-17 | 5000.00 | NULL    | 10     |
| 7844  | TURNER    | SALESMAN  | 7698 | 1981-09-08 | 1500.00 | 0.00    | 30     |
| 7876  | ADAMS     | CLERK     | 7788 | 1983-01-12 | 1100.00 | NULL    | 20     |
| 7900  | JAMES     | CLERK     | 7698 | 1981-12-03 | 950.00  | NULL    | 30     |
| 7902  | FORD      | ANALYST   | 7566 | 1981-12-03 | 3000.00 | NULL    | 20     |
| 7934  | MILLER    | CLERK     | 7782 | 1982-01-23 | 1300.00 | NULL    | 10     |

```
16 rows in set (0.00 sec)
```

```
mysql> select avg(sal) from emp where deptno=30;
```

| avg(sal)    |
|-------------|
| 1566.666667 |

```
1 row in set (0.00 sec)
```

```
mysql> select * from emp where sal > (select avg(sal) from emp where (select deptno from emp where ename='allen'));
```

| EMPNO | ENAME     | JOB       | MGR  | HIREDATE   | SAL     | COMM   | DEPTNO |
|-------|-----------|-----------|------|------------|---------|--------|--------|
| 7782  | CLARK     | MANAGER   | 7839 | 1981-06-09 | 2450.00 | NULL   | 10     |
| 7698  | BLAKE     | MANAGER   | 7839 | 1981-05-01 | 2850.00 | NULL   | 30     |
| 7566  | JONES     | MANAGER   | 7839 | 1981-04-02 | 2975.00 | NULL   | 20     |
| 7780  | jay_dbda  | analyst   | 7902 | 2024-10-08 | 3000.00 | 300.00 | 20     |
| 7781  | tanu_dbda | analyst   | 7902 | 2024-10-08 | 3000.00 | 300.00 | 20     |
| 7788  | SCOTT     | ANALYST   | 7566 | 1982-12-09 | 3000.00 | NULL   | 20     |
| 7902  | FORD      | ANALYST   | 7566 | 1981-12-03 | 3000.00 | NULL   | 20     |
| 7839  | KING      | PRESIDENT | NULL | 1981-11-17 | 5000.00 | NULL   | 10     |

```
8 rows in set (0.00 sec)
```

47. list all employees who are working in purchase department

48. list all employees who earn more than average salary of their own department

```
mysql> select * from emp e where sal > (select avg(sal) from emp m where e.deptno=m.deptno);
+-----+-----+-----+-----+-----+-----+-----+
| EMPNO | ENAME   | JOB       | MGR   | HIREDATE   | SAL      | COMM      | DEPTNO |
+-----+-----+-----+-----+-----+-----+-----+
| 7499 | ALLEN   | SALESMAN  | 7698  | 1981-02-20 | 1600.00  | 300.00    | 30     |
| 7566 | JONES   | MANAGER   | 7839  | 1981-04-02 | 2975.00  | NULL      | 20     |
| 7698 | BLAKE   | MANAGER   | 7839  | 1981-05-01 | 2850.00  | NULL      | 30     |
| 7780 | jay_dbda | analyst   | 7902  | 2024-10-08 | 3000.00  | 300.00    | 20     |
| 7781 | tanu_dbda | analyst   | 7902  | 2024-10-08 | 3000.00  | 300.00    | 20     |
| 7788 | SCOTT   | ANALYST   | 7566  | 1982-12-09 | 3000.00  | NULL      | 20     |
| 7839 | KING    | PRESIDENT | NULL   | 1981-11-17 | 5000.00  | NULL      | 10     |
| 7902 | FORD    | ANALYST   | 7566  | 1981-12-03 | 3000.00  | NULL      | 20     |
+-----+-----+-----+-----+-----+-----+-----+
8 rows in set (0.00 sec)

mysql> select avg(sal) from emp;
+-----+
| avg(sal) |
+-----+
| 2189.062500 |
+-----+
1 row in set (0.00 sec)

mysql> select avg(sal) from emp where deptno=30;
+-----+
| avg(sal) |
+-----+
| 1566.666667 |
+-----+
1 row in set (0.00 sec)

mysql> select avg(sal) from emp where deptno=20;
+-----+
| avg(sal) |
+-----+
| 2410.714286 |
+-----+
1 row in set (0.00 sec)
```

49. list all employees who earn sal < than their managers salary

```
mysql> select * from emp e where sal < (select sal from emp m where m.empno=e.mgr);
+-----+-----+-----+-----+-----+-----+-----+
| EMPNO | ENAME   | JOB       | MGR   | HIREDATE   | SAL      | COMM      | DEPTNO |
+-----+-----+-----+-----+-----+-----+-----+
| 7369 | SMITH   | CLERK     | 7902  | 1980-12-17 | 800.00   | NULL      | 20     |
| 7499 | ALLEN   | SALESMAN  | 7698  | 1981-02-20 | 1600.00  | 300.00    | 30     |
| 7521 | WARD    | SALESMAN  | 7698  | 1981-02-22 | 1250.00  | 500.00    | 30     |
| 7566 | JONES   | MANAGER   | 7839  | 1981-04-02 | 2975.00  | NULL      | 20     |
| 7654 | MARTIN  | SALESMAN  | 7698  | 1981-09-28 | 1250.00  | 1400.00   | 30     |
| 7698 | BLAKE   | MANAGER   | 7839  | 1981-05-01 | 2850.00  | NULL      | 30     |
| 7782 | CLARK   | MANAGER   | 7839  | 1981-06-09 | 2450.00  | NULL      | 10     |
| 7844 | TURNER  | SALESMAN  | 7698  | 1981-09-08 | 1500.00  | 0.00      | 30     |
| 7876 | ADAMS   | CLERK     | 7788  | 1983-01-12 | 1100.00  | NULL      | 20     |
| 7900 | JAMES   | CLERK     | 7698  | 1981-12-03 | 950.00   | NULL      | 30     |
| 7934 | MILLER  | CLERK     | 7782  | 1982-01-23 | 1300.00  | NULL      | 10     |
+-----+-----+-----+-----+-----+-----+-----+
11 rows in set (0.00 sec)
```

50. list all employees who are earning more than average salary of their job

51. display employee name and department

```
mysql> select ename,deptno from emp;
```

| ename     | deptno |
|-----------|--------|
| SMITH     | 20     |
| ALLEN     | 30     |
| WARD      | 30     |
| JONES     | 20     |
| MARTIN    | 30     |
| BLAKE     | 30     |
| jay_dbda  | 20     |
| tanu_dbda | 20     |
| CLARK     | 10     |
| SCOTT     | 20     |
| KING      | 10     |
| TURNER    | 30     |
| ADAMS     | 20     |
| JAMES     | 30     |
| FORD      | 20     |
| MILLER    | 10     |

```
16 rows in set (0.00 sec)
```

52. display empno,name,department name and grade (use emp,dept and salgrade table)

```
mysql> select e.empno,e.ename,d.dname,s.grade from emp e , dept d , salgrade s;
```

| empno | ename | dname      | grade |
|-------|-------|------------|-------|
| 7369  | SMITH | ACCOUNTING | 5     |
| 7369  | SMITH | RESEARCH   | 5     |
| 7369  | SMITH | SALES      | 5     |
| 7369  | SMITH | OPERATIONS | 5     |
| 7369  | SMITH | ACCOUNTING | 4     |
| 7369  | SMITH | RESEARCH   | 4     |
| 7369  | SMITH | SALES      | 4     |
| 7369  | SMITH | OPERATIONS | 4     |
| 7369  | SMITH | ACCOUNTING | 3     |
| 7369  | SMITH | RESEARCH   | 3     |
| 7369  | SMITH | SALES      | 3     |
| 7369  | SMITH | OPERATIONS | 3     |
| 7369  | SMITH | ACCOUNTING | 2     |
| 7369  | SMITH | RESEARCH   | 2     |
| 7369  | SMITH | SALES      | 2     |
| 7369  | SMITH | OPERATIONS | 2     |
| 7369  | SMITH | ACCOUNTING | 1     |
| 7369  | SMITH | RESEARCH   | 1     |
| 7369  | SMITH | SALES      | 1     |
| 7369  | SMITH | OPERATIONS | 1     |
| 7369  | SMITH | ACCOUNTING | 5     |
| 7369  | SMITH | RESEARCH   | 5     |
| 7369  | SMITH | SALES      | 5     |
| 7369  | SMITH | OPERATIONS | 5     |
| 7369  | SMITH | ACCOUNTING | 4     |
| 7369  | SMITH | RESEARCH   | 4     |

53. list all employees number,name, mgrno and manager name

```
mysql> select e.empno,e.ename,e.mgr,m.ename from emp e inner join emp m on m.empno=e.mgr;
```

| empno | ename  | mgr  | ename |
|-------|--------|------|-------|
| 7369  | SMITH  | 7902 | FORD  |
| 7499  | ALLEN  | 7698 | BLAKE |
| 7521  | WARD   | 7698 | BLAKE |
| 7566  | JONES  | 7839 | KING  |
| 7654  | MARTIN | 7698 | BLAKE |
| 7698  | BLAKE  | 7839 | KING  |
| 7782  | CLARK  | 7839 | KING  |
| 7788  | SCOTT  | 7566 | JONES |
| 7844  | TURNER | 7698 | BLAKE |
| 7876  | ADAMS  | 7788 | SCOTT |
| 7900  | JAMES  | 7698 | BLAKE |
| 7902  | FORD   | 7566 | JONES |
| 7934  | MILLER | 7782 | CLARK |

13 rows in set (0.03 sec)

54. create following tables and solve following questions(primary keys are marked in yellow)  
foreign keys are marked in green

product(pid,pname,price,qty,cid,sid)

salesman (sid,sname,address)

1. list all product name,their category name and name of a person, who sold that product
2. list all product name and salesman name for all salesman who stays in pune

3. list all product name and category name

55. create following tables and solve following questions(primary keys are marked in yellow)  
foreign keys are marked in green

faculty(fid,fname,sp.skill1,sp.skill2)

```
mysql> create table faculty(  
-> fid int primary key,  
-> fname varchar(20),  
-> skill1 varchar(20),  
-> skill2 varchar(20)  
-> );  
Query OK, 0 rows affected (0.27 sec)
```

courses(cid,cname,rid,fid)

room(roomid,rname,rloc)

```
mysql> create table room(  
-> rid int primary key,  
-> rname varchar(20),  
-> rloc varchar(20)  
-> );  
Query OK, 0 rows affected (0.26 sec)
```

faculty

fid fname spskill1 spskill2

10 kjzhcjh a b

```
mysql> insert into faculty values(10,'aaa','a','b');
Query OK, 1 row affected (0.05 sec)
```

11 sdd x z

```
mysql> insert into faculty values(11,'bbb','x','z');
Query OK, 1 row affected (0.04 sec)
```

12 lksjk a x

```
mysql> insert into faculty values(12,'ccc','a','x');
Query OK, 1 row affected (0.05 sec)
```

13 ksdjlkj a b

```
mysql> insert into faculty values(13,'ddd','a','b');
Query OK, 1 row affected (0.06 sec)
```

```
mysql> select * from faculty;
+-----+-----+-----+-----+
| fid | fname | skill1 | skill2 |
+-----+-----+-----+-----+
| 10 | aaa | a | b |
| 11 | bbb | x | z |
| 12 | ccc | a | x |
| 13 | ddd | a | b |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

courses

cid cname rid fid

121 DBDA 100 10

131 DAC 101

141 DTISS

151 DIOT 105 12

Room

roomid rname rloc

100 jasmin 1st floor

```
mysql> insert into room values(100,'jasmin','1st floor');
Query OK, 1 row affected (0.04 sec)
```

101 Rose 2nd floor

```
mysql> insert into room values(101,'rose','2nd floor');
Query OK, 1 row affected (0.04 sec)
```

105 Lotus 1st floor

```
mysql> insert into room values(105,'lotus','1st floor');
Query OK, 1 row affected (0.04 sec)
```

103 Mogra 1st floor

```
mysql> insert into room values(103,'mogra','1st floor');
Query OK, 1 row affected (0.03 sec)
```

```
mysql> select * from room;
+----+-----+-----+
| rid | rname | rloc  |
+----+-----+-----+
| 100 | jasmin | 1st floor |
| 101 | rose   | 2nd floor |
| 103 | mogra  | 1st floor |
| 105 | lotus  | 1st floor |
+----+-----+-----+
4 rows in set (0.00 sec)
```

1. list all courses for which no room is assigned and all rooms for which are available
  2. list all faculties who are not allocated to any course and rooms which are not allocated to any course
  3. list all rooms which are allocated or not allocated to any courses
  4. list all rooms which are not allocated to any courses
  5. display courses and faculty assigned to those courses whose special skill is database
  6. display time table --- it should contain course details , faculty and room details
56. create following tables with given constraints
- product---- qty >0, default 20.00,pname not null and unique
- prodid pname qty price catid sid

123 lays 30 30.00 1 12

111 pepsi 40 50.00 4 11

134 nachos 50 50.00 1 12

124 dairy milk 40 60.00 2 14

124 pringles 40 60.00 1 14

saleman ----- sname -----not null

sid sname city

11 Rahul Pune

12 Kirti Mumbai

13 Prasad Nashik

14 Arnav Amaravati

category ----- cname unique and not null

cid cname description

1 chips very crunchy

2 chocolate very chocolaty

3 snacks yummy

4 cold drinks thanda thanda cool cool

1. List all products with category chips
2. display all products sold by kirti
3. display all salesman who do not sold any product
4. display all category for which no product is there
5. display all products with no category assigned
6. list all salesman who stays in city with name starts with P or N
7. add new column in salesman table by name credit limit

-----